

Since this is regression, I am using regression related classes only.

SVM:

Class	Kernel	Gamma	R2_Score
SVR	Rbf(default)	Scale(default)	-0.0573
SVR	Rbf(default)	auto	-0.0574
SVR	linear	auto	0.8774
SVR	linear	Scale(default)	0.8774
SVR	poly	Scale(default)	-0.0508
SVR	poly	auto	Kernel restart
SVR	sigmoid	auto	-0.0574
SVR	sigmoid	Scale(default)	-0.0575
SVR	precomputed	Scale(default)	Must be square matrix. Here input is 35*5
NuSVR	Rbf(default)	Scale(default)	-0.0405
NuSVR	Rbf(default)	auto	-0.0407
NuSVR	linear	auto	0.9317
NuSVR	linear	Scale(default)	Kernel restart
NuSVR	poly	Scale(default)	-0.0346
NuSVR	poly	auto	-13.9843
NuSVR	sigmoid	auto	-0.0407
NuSVR	sigmoid	Scale(default)	-0.0407
NuSVR	precomputed	Scale(default)	Must be square matrix. Here input is 35*5

Class	penalty	loss	R2_Score
LinearSVR	-	epsilon_insensitive(default)	0.5793
LinearSVR	-	squared_epsilon_insensitive	-1.7936

From the above, Best R2 score value is 0.9317. Class: NuSVR, kernel: linear, gamma = auto.

Decision Tree:

Class	criterion	splitter	R2_Score
DecisionTreeRegressor	Squared_error(default)	best(default)	0.9361
DecisionTreeRegressor	Squared_error(default)	random	0.5541
DecisionTreeRegressor	Friedman_mse	best(default)	0.9176
DecisionTreeRegressor	Friedman_mse	random	0.6423
DecisionTreeRegressor	Absolute_error	best(default)	0.9444
DecisionTreeRegressor	Absolute_error	random	0.9188
DecisionTreeRegressor	poisson	best(default)	0.9060
DecisionTreeRegressor	poisson	random	0.7378
ExtraTreeRegressor	Squared_error(default)	random (default)	0.8795
ExtraTreeRegressor	Squared_error(default)	best	0.9279
ExtraTreeRegressor	Friedman_mse	random (default)	0.9127
ExtraTreeRegressor	Friedman_mse	best	0.9045
ExtraTreeRegressor	Absolute_error	random (default)	0.8495
ExtraTreeRegressor	Absolute_error	best	0.9564
ExtraTreeRegressor	poisson	random (default)	0.9024
ExtraTreeRegressor	poisson	best	0.9079

From the above, Best R2 score value is 0.9564. Class: ExtraTreeRegressor, criterion: absolute_error, splitter = best.